Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/10/2026
PI Name:	Langen, Britta Ph.D.		
Project Title:	Sex-Specific Biomarker Identification for Space Radiation-Induced Lymphoma		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis (2) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020-2021 HERO 80JSC020N0001-HHCSR, Omnibus2. Human Health Countermeasures and Space Radiation Topics Appendix C; OMNIBUS2-Appendix D
Start Date:	02/01/2024	**End Date:**	08/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2025 per JSC Grants Office and NSSC documentation (Ed., 7/7/25).		
Key Personnel Changes/Previous PI:	The PI %Effort was changed from 20% to 80% due to the unexpected departure of the hosting lab for the PI's visiting appointment.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K0576		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The following major goals and scientific objectives were revised and approved by NASA SRE before the start of the project. The changes to the study design were made in response to %Effort constraints resulting from the unexpected departure of the hosting lab. Original major goals: - Identification of predictive biomarkers signatures for SimGCRSim induced lymphoma - Male- and female-specific radiation signatures - Signature robustness for diagnosis under mission parameters Retained major goals: - Male- and female-specific radiation signatures - Signature robustness for diagnosis under mission parameters Original aims: Aim 1: To identify blood-based sex-specific protein biomarker candidates for space radiation-induced lymphoma Aim 2a: To quantify sex-specific differences on the proteomic scale in blood plasma Aim 2b: and deduce sex-specific dependencies for risk projection Aim 3: To perform a robustness analysis of the biomarker candidate signatures for the space setting and assess confounding effects by circadian disruption Retained aims: Aim 2a: To quantify sex-specific differences on the proteomic scale in blood plasma Aim 3: To perform a robustness analysis of the biomarker candidate signatures for the space setting and assess confounding effects by circadian disruption Revised aims: Aim 2b: To assess sex effects on established lymphoma induction pathways NASA Gaps: Retained Gaps: CANCER-301: Determine the tissue-specific contributions to overall space radiation carcinogenic risk and whether there are sex-specific dependencies SLEEP-101: Given each crew member will experience multiple spaceflight hazards simultaneously, we need to identify and characterize the potential additive, antagonistic, or synergistic impacts of multiple stressors (e.g., Space Radiation, Altered Gravity, Isolation, altered immune, altered sleep) on crew sleep-wake cycles and/or circadian shifting, health and/or CNS/cognitive functioning to identify any identified adverse individual or team crew health, and/or operationally-relevant performance outcomes. CANCER-101: Identify the combined effects of spaceflight stressors on identified surrogate cancer endpoints and shared biology with other degenerative diseases. Discarded Gaps CANCER-401: Identify biomarkers for estimating individual susceptibility, risk assessment, and health monitoring. Technical approach: The irradiation procedure was revised to accommodate the compressed timeline. Animals were irradiated with protons at MD Anderson, Houston, TX, instead of SimGCRSim at the Brookhaven National Laboratory, Upton, NY. Male and female 10-weeks-old BALB/C mice were subjected to total body proton irradiation as an approximation to simulated space radiation. Following irradiation, animals were housed under standard light conditions or subjected to a simulated shift work regimen shifted by 4 hours. Three weeks post irradiation, the animals were killed under anesthesia to collect blood plasma and several solid tissues with relevance to the circadian axis and blood formation. Samples were committed to next-generation proteomics to quantitate radiation-induced changes in protein abundance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Molecular biomarkers are a necessary tool for radiation risk assessment and for early diagnosis and intervention of space radiation-induced malignancies. Blood-based protein biomarkers in particular are minimally invasive, robust in analytical processing, and easy to implement for onsite monitoring. The ultimate goal of biomarker assays is to increase astronaut safety during missions to the Moon and Mars and improve long-term health upon return. This project investigates the impact of circadian disruption on the radiation dose-effect relationship across the proteome under consideration of sexual dimorphism. The goal is to develop sex-specific biomarker assays that are chronorobust, meaning accurate irrespective of sleep-wake rhythms. The overall benefits of this effort are the 1) promotion of sex equity in radiation risk assessment and 2) advancement of personalized medicine with regard to occupational or lifestyle habits.
Task Progress:	1) Major activities The animal cohort consisted of 28 male and 28 female BALB/C mice aged to 10-weeks at irradiation (Day 0). The female cohort was selected for metestrus/diestrus phase from 60 females. Fourteen animals (per sex) were total-body irradiated with 1.0 Gy absorbed dose from proton beams (MD Anderson, Houston, TX), and the other half was mock-treated, including anesthesia. Treatments were performed between 21.14 PM and 23.39 PM. On Day 2 post irradiation, the animals were transported to UT Southwestern Medical Center, Dallas, TX, and subjected to quarantine. All animals cleared expedited quarantine on Day 14. The cohort was divided into two arms: half the cohort was housed under standard 6 AM/6 PM light-dark (LD) cycle for the duration of the study (3 weeks post irradiation), and the other half was subjected to simulated shift work, i.e., housed for one week in 10 AM/10 PM LD cycle in circadian cabinets every other week until the end of the study. Cages were moved during matching dark phases. Animals were sacrificed under anesthesia (isoflurane overdose via inhalation) and whole blood, the thyroid, the left kidney, and left femur were collected. The cohort was separated into four groups and sacrificed between approximately 9:30 AM and 12 PM on four consecutive days to ensure congruent circadian times. Solid organs were flash-frozen in liquid nitrogen upon excision. Whole blood was collected in heparin-treated syringes and centrifuged to separate plasma. Plasma was frozen and serum was discarded. Samples were stored at -80°C until further processing. Kidney medulla samples were dissected on dry ice. Blood plasma, kidney medulla, and thyroid were submitted to high-throughput protein mass spectrometry. Analysis is ongoing at the time of writing. 2) Specific objectives (aims)

	All animal and sample collection procedures have been completed. Sample processing and analysis are ongoing at the time of writing. All revised Aims and Gaps are planned to be met. Please note that due to the administrative delays, the planned study duration of 2 months was shortened to 3 weeks to accommodate the PI's visa constraints and departure from the United States. 3) Significant results or key outcomes Sample analysis is pending at the time of writing. Significant results and conclusions will be reported in scientific research papers. Grants officials will be made aware of any publications resulting from this effort.
Bibliography Type:	Description: (Last Updated:)